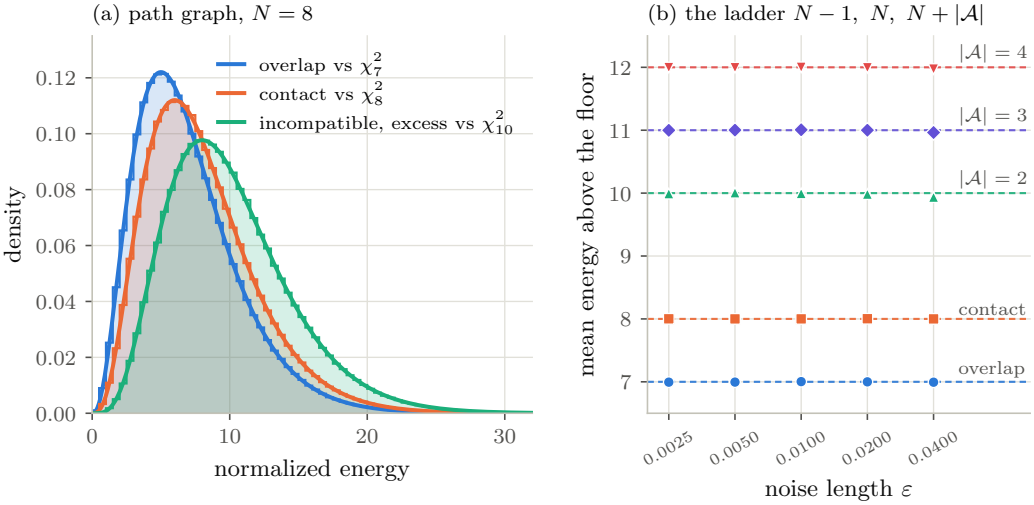

(a) path graph, $N = 8$
overlap vs χ^2_7
contact vs χ^2_8
incompatible, excess vs χ^2_{10}
density
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
10
20
30
normalized energy
(b) the ladder $N-1,\ N,\ N+|\mathcal{A}|$
mean energy above the floor
12
11
10
9
8
7
$|\mathcal{A}| = 4$
$|\mathcal{A}| = 3$
$|\mathcal{A}| = 2$
contact
overlap
0.0025
0.0050
0.0100
0.0200
0.0400
noise length ε